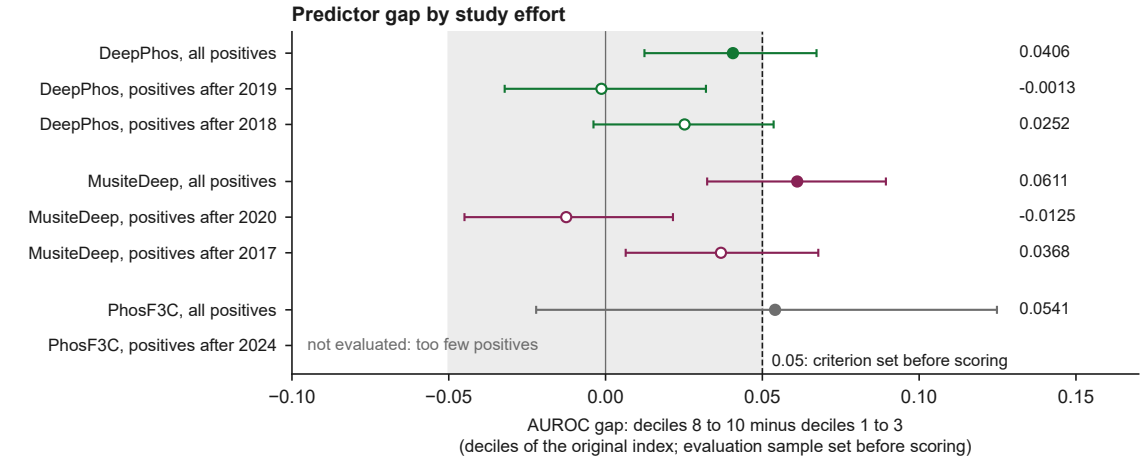
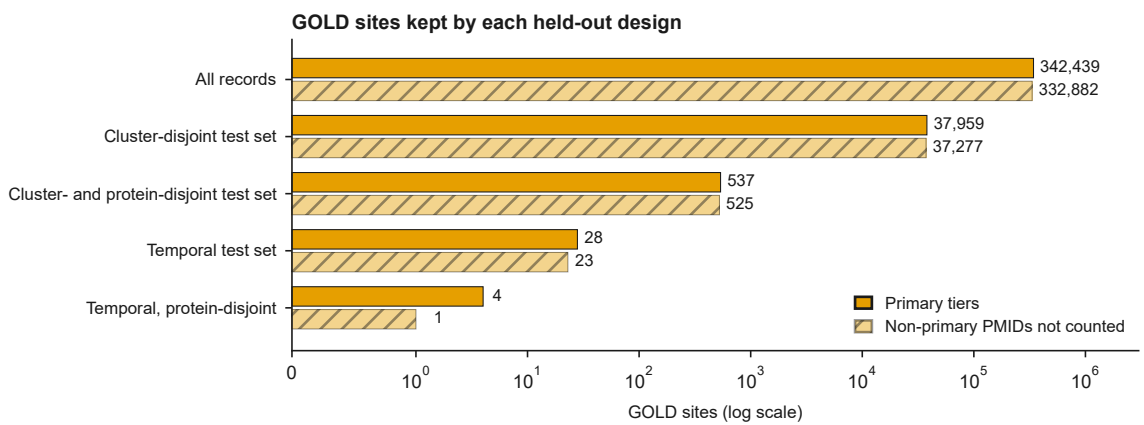
A



B



Secondary analysis. A: serine/threonine sites for DeepPhos and MusiteDeep, tyrosine sites for PhosF3C; filled marker with the 95% gene-cluster interval for all positives, open marker with the 90% interval for positives first reported after the stated year; primary restriction year per tool: DeepPhos 2019, MusiteDeep 2020, PhosF3C 2024 (year of the first publication describing the version scored); the other restriction year shown for DeepPhos and MusiteDeep is a sensitivity analysis; grey band: equivalence margin; sites, positives and gene clusters: DeepPhos all: 8,641 sites, 4,485 positives, 5,209 gene clusters; DeepPhos 2019: 5,383 sites, 1,227 positives, 3,561 gene clusters; DeepPhos 2018: 5,780 sites, 1,624 positives, 3,696 gene clusters; MusiteDeep all: 8,641 sites, 4,485 positives, 5,209 gene clusters; MusiteDeep 2020: 5,124 sites, 968 positives, 3,435 gene clusters; MusiteDeep 2017: 5,890 sites, 1,734 positives, 3,767 gene clusters; PhosF3C all: 1,359 sites, 515 positives, 1,218 gene clusters. B: set-B tiers from the study-cluster rule and tiers with the non-primary PMIDs not counted; test sets as assigned before scoring. Designs with no GOLD site are not drawn: temporal, study-disjoint (test set of 411 sites); temporal, protein- and study-disjoint (test set of 47 sites).